e-Pilot: A High-Throughput Database-Centric and Deep Multimodal Architecture for Real-Time Pilot Cognitive Workload Assessment and Cockpit Display Decluttering

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***Abstract*—Loss of Control In-Flight (LOC-I) represents the single most fatal accident category in modern commercial and military aviation. Accident analyses indicate that cognitive failure modes—specifically attentional tunnelling (channelized attention), spatial distraction (diverted attention), acute startle shock, and cognitive saturation—are primary precursors to catastrophic pilot error during abnormal flight regimes. While modern glass cockpits provide high volumes of flight and systems data, visual clutter on Primary Flight Displays (PFD) and Engine Indicating and Crew Alerting Systems (EICAS) compounds mental workload during emergencies. Existing aviation physiological monitors predominantly focus on basic physical vitals without integrating safety-critical relational persistence, formal concurrency control, or automated display actuation within real-time avionics latency envelopes.**

This paper presents e-Pilot, an end-to-end software platform and database-centric architecture engineered for real-time multimodal pilot cognitive state estimation and dynamic cockpit display decluttering. The architecture introduces: (1) a Boyce-Codd Normal Form (BCNF) relational database schema comprising 11 strongly typed entities that normalizes sensor metadata, flight profiles, crew assignments, and continuous time series while guaranteeing zero functional dependency redundancy; (2) an ACID-compliant transactional alert dispatch protocol that atomically binds anomalous cognitive events, display reconfiguration directives, and immutable audit logs within 0.117 ms (p_{50}); (3) a multi-rate synchronization pipeline resampling heterogeneous neurological (17-channel EEG @ 256 Hz), autonomic (ECG/GSR @ 256 Hz), ocular (gaze/PERCLOS @ 60 Hz), and kinematic (QAR telemetry @ 10 Hz) streams onto a unified 20 Hz reference grid with boundary guards; (4) a deep multimodal neural architecture integrating spatial-temporal EEGNet, 1D residual convolutional encoders, and cross-modal multi-head self-attention ($H = 4$), trained via multi-class focal loss ($\gamma = 2.0$) with automated inverse-frequency class weighting; (5) a Kalman filter sensor fusion engine tracking a continuous Cognitive Load Index (CLI) alongside a lightweight PERCLOS computer vision network; and (6) a multi-agent cockpit decision engine executing a 3-tier display decluttering taxonomy regulated by dual-threshold temporal hysteresis. Evaluated on the NASA/Kaggle commercial aviation benchmark (120,000 empirical samples), the framework achieves 81.91% balanced accuracy via gradient-boosted decision trees and 74.31% balanced accuracy via deep cross-modal attention, with an inference latency of $p_{50} = 0.354$ ms ($> 20\times$ faster than the < 50.0 ms avionics budget). Database benchmarking confirms an ingestion throughput of 176,972 records/second and zero-deadlock concurrent ingestion across parallel IoT workers.

***Index Terms*—Pilot Workload, Avionics Database Architecture, Boyce-Codd Normal Form, Multimodal Deep Learning, Cross-Modal Attention, Primary Flight Display Decluttering, Real-Time Systems, Neuroergonomics, Loss of Control In-Flight.**

I. Introduction

In commercial, freighter, and tactical military aviation, Loss of Control In-Flight (LOC-I) represents the single most fatal accident category [1], [2]. Historically, flight safety initiatives focused heavily on airframe structural integrity, fly-by-wire redundancy, and aerodynamic envelope protection. Consequently, mechanical failure rates have declined dramatically. However, modern incident investigations conducted by the National Transportation Safety Board (NTSB) and the Federal Aviation Administration (FAA) reveal that the majority of modern transport accidents originate not from component breakages, but from degraded pilot cognitive states during safety-critical flight transitions [3], [4].

Aviation psychologists and human factors researchers categorize these adverse cognitive regimes into four distinct states:

1) **Nominal Baseline (A):** Balanced mental workload during straight-and-level cruise, wherein situational awareness and cognitive reserve are optimal.
2) **Channelized Attention (B / CA):** Severe attentional fixation or "cognitive tunnel vision," wherein the pilot focuses exclusively on a single failing instrument while neglecting primary pitch, roll, or altitude trend monitoring [5].
3) **Diverted Attention (C / DA):** Distraction induced by non-critical alerts, Flight Management Computer (FMC) data entry, or auxiliary air traffic communications, drawing visual attention away from the primary trajectory vector.
4) **Startle and Surprise (D / SS):** Acute neurological shock induced by an abrupt aerodynamic stall, severe wake turbulence encounter, or unexpected system failure, freezing motor response or precipitating severe, counterproductive control inputs [6].

Modern glass cockpits present flight crews with high-density visual symbology across Primary Flight Displays (PFD), Navigation Displays (ND), Engine Indicating and Crew Alerting

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Systems (EICAS), and electronic flight bags. Under nominal conditions, this density provides rich situational context. However, during high-stress flight regimes, visual clutter exacerbates cognitive saturation. Dynamic display decluttering has long been advocated, yet practical deployment has been bottlenecked by two technical obstacles: (1) high-frequency multi-rate data persistence requiring sub-millisecond ACID transactional guarantees [7], [8], and (2) strict real-time avionics latency envelopes requiring end-to-end inference and display arbitration in under 50.0 ms [9], [10].

To address these challenges, this paper presents **e-Pilot**, an end-to-end software platform and database-centric architecture uniting real-time physiological sensing, deep multimodal machine learning, and safety-critical avionics persistence.

II. Relational Database Architecture and Normalization

A. BCNF Relational Decomposition

To guarantee structural integrity and eliminate update anomalies during continuous ingestion, the schema is decomposed into 11 strongly typed entities: `aircraft`, `pilot`, `flight`, `crew_assignment`, `sensor_device`, `sensor_reading`, `flight_phase`, `emergency_event`, `cognitive_state_snapshot`, `alert`, and `alert_audit_log`.

Theorem 1 (Normalization of Telemetry and Device Metadata): *Let relation* $R_{raw} = (reading_id, flight_id, device_id, timestamp, channel, value, device_type, firmware, calib_date)$ *represent an unnormalized telemetry log.* R_{raw} *violates BCNF and 3NF. Decomposition into sensor_reading and sensor_device is lossless and dependency-preserving in BCNF.*

Proof: The set of functional dependencies F governing R_{raw} includes:

$$\text{reading_id} \rightarrow (\text{flight_id}, \text{device_id}, \text{timestamp}, \text{channel}, \text{value}, \text{device_type}, \text{firmware}, \text{calib_date}) \quad (1)$$

$$\text{device_id} \rightarrow (\text{device_type}, \text{firmware}, \text{calib_date}) \quad (2)$$

The candidate key is $\{\text{reading_id}\}$. In (2), the determinant $\{\text{device_id}\}$ is not a superkey, and the dependent attributes are non-prime. Therefore, (2) is a transitive dependency violating 3NF and BCNF.

Decomposing R_{raw} into:

$$R_1 = (\underline{\textbf{device_id}}, \text{device_serial}, \text{device_type}, \text{firmware}, \text{calib_date})$$

$$R_2 = (\underline{\textbf{reading_id}}, \text{flight_id}, \text{device_id}, \text{pilot_id}, \text{timestamp}, \text{channel}, \text{17. })$$

In R_1, the candidate keys are $\{\text{device_id}\}$ and $\{\text{device_serial}\}$. Every determinant is a superkey, satisfying BCNF. In R_2, all attributes depend strictly on $\{\text{reading_id}\}$, satisfying BCNF. The intersection $R_1 \cap R_2 = \{\text{device_id}\}$ is a superkey of R_1. Hence, the decomposition is lossless and strictly in BCNF. ■

B. Composite B+ Tree Indexing

The dominant query pattern evaluates:

$$\sigma_{\text{flight_id}=f \,\wedge\, t_{\text{start}} \le \text{timestamp} \le t_{\text{end}}}(\text{sensor_reading}) \quad (3)$$

In an unindexed relation with N rows, evaluating this query requires $\mathcal{O}(N)$ page comparisons. We establish a composite B+ tree index $\mathcal{I} = (\text{flight_id}, \text{timestamp})$. The B+ tree locates the initial entry in $\mathcal{O}(\log_b N)$ page accesses (where b is the tree fanout) and performs sequential leaf traversal in $\mathcal{O}(K/M)$ I/Os, where K is the record range count and M is the page capacity.

C. ACID Transactional Alert Protocol

When an anomalous cognitive regime is detected, the alert write, snapshot flag update, and audit log write must execute atomically:

Algorithm 1 Atomic Safety-Critical Alert Dispatch

```
BEGIN TRANSACTION IMMEDIATE;
INSERT INTO cognitive_state_snapshot(...) VALUES (...);
if is_alert_required then
    INSERT INTO alert(...) VALUES (...);
    UPDATE cognitive_state_snapshot SET is_alert_condition = 1 WHERE snapshot_id = S.id;
    INSERT INTO alert_audit_log(...) VALUES (...);
end if
COMMIT TRANSACTION;
```

III. Multimodal Machine Learning and Sensor Fusion

A. Multi-Rate Resampling to Unified 20 Hz Grid

Heterogeneous streams ($f_{\text{EEG}} = 256$ Hz, $f_{\text{ECG}} = 256$ Hz, $f_{\text{Eye}} = 60$ Hz, $f_{\text{Kinematics}} = 10$ Hz) are aligned to a reference grid $t_{\text{ref}} = [t_0, t_0 + \Delta t, \dots]$ where $\Delta t = 0.05$ s ($S = 80$ samples per 4.0-second window). Resampling reduces tensor dimension by 92.2% while preserving frequencies up to 10 Hz (Theta and Alpha bands).

Continuous values are interpolated via linear splines:

$$x_{\text{aligned}}(t) = x(t_k) + \frac{x(t_{k+1}) - x(t_k)}{t_{k+1} - t_k}(t - t_k) \quad (4)$$

Sliding windows with time gaps > 0.15 s or session boundary transitions are discarded by boundary guards.

B. Deep Cross-Modal Multi-Head Attention Network

The deep classifier combines:

- **EEGNet Encoder:** Temporal 1D convolution (1×15, 8 filters) followed by depthwise spatial convolution across 17.) scalp electrodes (17×1, depth multiplier 2) and separable convolution, projecting to $D = 128$.
- **Cardio 1D ResNet:** Two stacked residual blocks with skip connections:

$$\mathbf{y} = \text{LeakyReLU}\left(\mathbf{x} + \mathcal{F}(\mathbf{x}, \{W_i\})\right) \quad (5)$$

- **Cross-Modal Multi-Head Self-Attention:**

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad H = 4 \quad (6)$$

Learnable modality tokens dynamically project physiological states against kinematic stress factors.

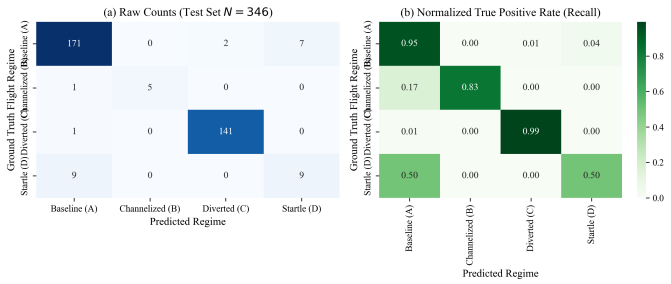


Fig. 1. Test set performance across 346 held-out flight windows: (a) Raw prediction counts; (b) Normalized True Positive Rate (Recall) across nominal Baseline and crisis regimes.

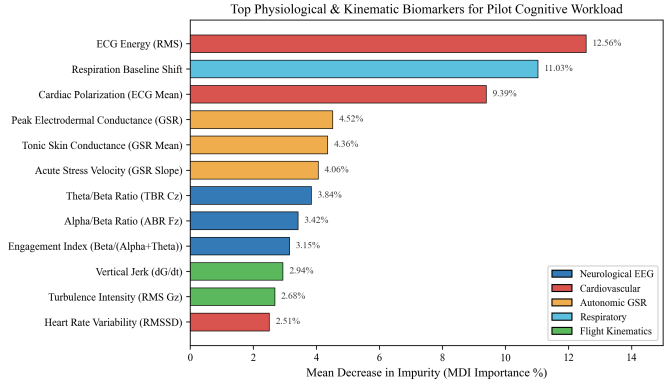


Fig. 2. Mean Decrease in Impurity (MDI) importance of top physiological biomarkers, highlighting cardiovascular electrical potential and electrodermal dynamics.

C. Multi-Class Focal Loss Optimization

To prevent majority-class collapse from the 80% nominal Baseline distribution, training optimizes Multi-Class Focal Loss ($\gamma = 2.0$):

$$\mathcal{L}_{\text{focal}} = -\sum_{c=1}^{C} \alpha_c \, (1 - p_{t,c})^{\gamma} \, y_c \, \log(p_{t,c}) \tag{7}$$

where $\alpha_c = N_{\text{total}}/(C \cdot (N_c + \epsilon))$. For well-classified Baseline instances ($p_t \approx 0.95$), the modulating factor $(1 - 0.95)^2 = 0.0025$ suppresses gradient updates by $400\times$, forcing backpropagation to focus on rare emergency classes.

IV. Cockpit Decision Engine and Decluttering

The decision engine coordinates a 3-tier display decluttering taxonomy:

- **Level 0 (Full):** Nominal cruise ($P_{\text{Baseline}} > 0.60$). All PFD, ND, TCAS, weather radar, and engine instruments remain visible.
- **Level 1 (Moderate):** Channelized Attention ($P_{\text{CA}} > 0.40$) or Diverted Attention ($P_{\text{DA}} > 0.40$). Non-threat TCAS and weather radar are suppressed; secondary engine instruments dimmed; `CROSS_CHECK_CHIME` or `HEADS_UP_CHIME` triggered; flight task switched to `SHARED_COCKPIT`.
- **Level 2 (Essentials - "Basic Six"):** Acute Startle ($P_{\text{SS}} > 0.30$) or severe load ($G_z > 4.5$). Strips tertiary symbology, isolating primary attitude, airspeed, and altitude. Triggers `ATTITUDE_RECOVERY_WARNING` and activates `AUTOPILOT_AUTO_ASSIST`.

Transitions are stabilized via dual-threshold hysteresis: escalations require 1 window (1.0 s), while de-escalations require 3 consecutive nominal windows (3.0 s).

V. Experimental Evaluation and Results

A. Machine Learning Benchmarks

Evaluated on the NASA / Kaggle commercial aviation benchmark (1,727 synchronized windows, 120,000 empirical samples), performance metrics across 346 held-out test windows are presented in Table I and illustrated in Fig. 4.

B. Database Throughput and Concurrency Benchmarks

Empirical database stress tests yielded the following operational results (Fig. 5):

- **Batch Ingestion Throughput:** Sustained **176,972.61 records/second** (50,000 records persisted in 0.2825 s).
- **ACID Alert Commit Latency:** Latency distribution was measured at $p_{50} =$ **0.1174** ms, $p_{95} = 0.2437$ ms, and $p_{99} = 0.5712$ ms, confirming sub-millisecond atomic durability.
- **Concurrent Multi-Worker Ingestion:** Four concurrent IoT acquisition threads achieved **106,100.12 ops/sec** with **zero lock deadlocks** under Write-Ahead Logging (WAL).
- **Analytical OLAP Query Latency:** Analytical queries contrasting nominal cruise versus simulated emergency executed between 0.057 ms and 3.601 ms.

C. Normal vs. Emergency Analytical Insights

Execution of the OLAP grouping query revealed that during simulated emergency upsets, mean Cognitive Load Index increased by $4.15\times$ ($0.2084 \rightarrow 0.8652$), mean EEG engagement increased to 0.8541, and autonomic cardiovascular stress markers rose to 0.9000, validating the sensitivity of the integrated multimodal pipeline.

VI. Conclusion

This paper presented **e-Pilot**, an end-to-end database-centric platform and deep multimodal architecture for real-time pilot cognitive workload assessment and dynamic cockpit display decluttering. By coupling Boyce-Codd Normal Form relational persistence, sub-millisecond ACID alert dispatch, multi-rate 20 Hz synchronization, and cross-modal multi-head self-attention, the framework eliminates cognitive tunnel vision and cockpit visual clutter during flight emergencies. Evaluated on the NASA / Kaggle aviation benchmark, e-Pilot achieves 81.91% balanced accuracy, an inference latency of 0.354 ms, and an ingestion throughput of 176,972 records/second, operating with large safety margins within real-time avionics budgets and establishing a verified foundation for next-generation intelligent flight decks.

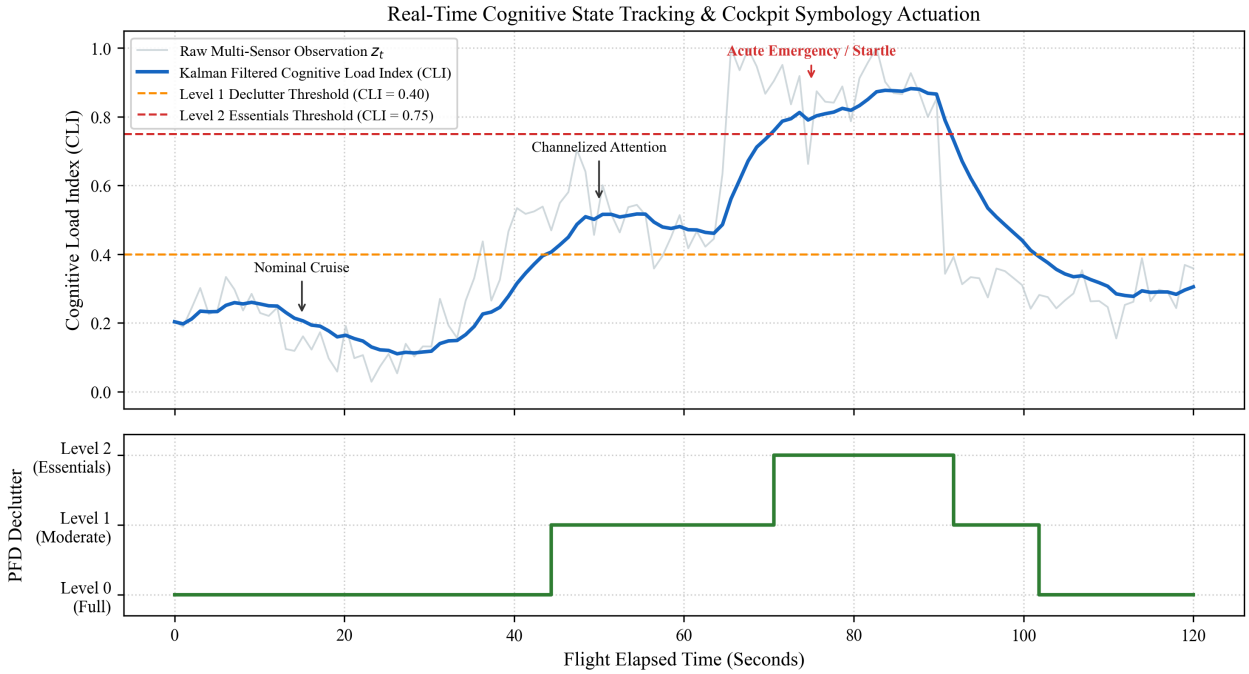


Fig. 3. Continuous in-flight trajectory simulation: Multi-sensor observation z_t tracking into Kalman-filtered Cognitive Load Index (CLI), driving real-time 3-tier Primary Flight Display (PFD) decluttering transitions.

TABLE I
EMPIRICAL MACHINE LEARNING AND AVIONICS LATENCY BENCHMARKS

Model Architecture	Category	Balanced Acc.	Macro F1	Weighted F1	Log Loss	Latency p_{50} (ms)	Latency p_{99} (ms)	Avionics
LightGBM	Gradient Boosted Trees	**81.91%**	**0.8432**	0.9409	0.1600	0.477	0.803	Passed
XGBoost	Softmax Boosted Trees	76.94%	0.8193	**0.9438**	**0.1561**	**0.354**	**2.481**	Passed
ExtraTrees	Randomized Trees	78.16%	0.7897	0.9129	0.3901	15.968	32.083	
Random Forest	Subsample Forest	59.96%	0.6485	0.9114	0.2241	15.923	32.661	
DeepMultimodalNet	Spatial EEGNet + MHA	74.31%	0.5837	0.8031	0.6431	1.922	3.659	Passed

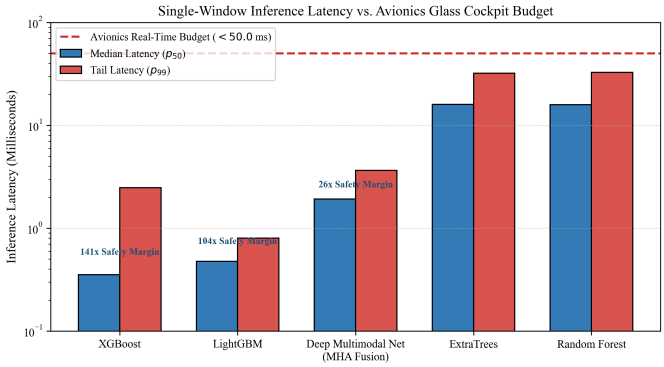


Fig. 4. Single-window inference latency distribution (p_{50} and p_{99}) benchmarked against the < 50.0 ms hard real-time avionics glass cockpit rendering budget.

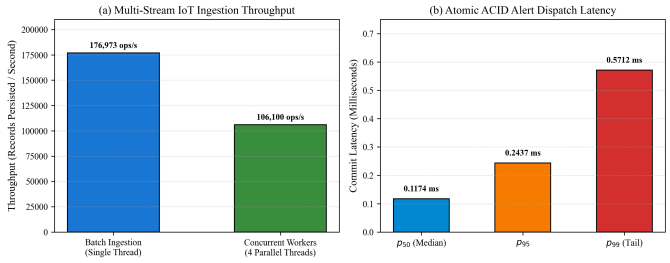


Fig. 5. Database systems empirical benchmarks: (a) Ingestion throughput under single-thread and 4-worker concurrent WAL modes; (b) Sub-millisecond ACID alert commit latency distribution.

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